

DANESHVAR AMROLLAHI

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EDUCATION

- **Stanford University** 2024/01 - Present
PhD in Computer Science Advisor: Clark Barrett
- **Stanford University** 2024/01 - 2025/06
MSc in Computer Science
- **University of Tehran** 2018/09 - 2023/02
BSc in Computer Engineering (Software) GPA: 18.02/20.00

PUBLICATIONS

- C. Sun, Y. Sun, D. Amrollahi, E. Zhang, S. Lahiri, S. Lu, D. Dill, C. Barrett (2025). **VeriStruct: AI-assisted Automated Verification of Data-Structure Modules in Verus**. *Under Submission*.
- D. Amrollahi, M. Preiner, A. Niemetz, A. Reynolds, M. Charikar, C. Tinelli, C. Barrett (2025). **Towards SMT Solver Stability via Input Normalization**. *Formal Methods in Computer-Aided Design (FMCAD 2025)*.
- P. Hozzová, D. Amrollahi, M. Hajdu, L. Kovács, A. Voronkov, E.M. Wagner (2024). **Synthesis of Recursive Programs in Saturation**. *International Joint Conference on Automated Reasoning (IJCAR 2024)*.
- D. Amrollahi, E. Bartocci, G. Kenison, L. Kovács, M. Moosbrugger, M. Stankovič (2024). **(Un)Solvable Loop Analysis**. *Formal Methods in System Design (FMSD)*.
- D. Amrollahi, H. Hojjat, P. Rümmer (2024). **An Encoding for CLP Problems in SMT-LIB**. *10th Workshop on Horn Clauses for Verification and Synthesis (HCVS 2023)*.
- D. Amrollahi, E. Bartocci, G. Kenison, L. Kovács, M. Moosbrugger, M. Stankovič (2022). **Solving Invariant Generation for Unsolvable Loops**. *29th International Static Analysis Symposium (SAS 2022)*. **Awarded the Radhia Cousot Young Researcher Best Paper Award**.
- A. Humenberger, D. Amrollahi, N. Bjørner, L. Kovács (2022). **Algebra-Based Reasoning for Loop Synthesis**. *Formal Aspects of Computing (FAC)*.

INDUSTRY EXPERIENCE

- **Applied Science Intern at Amazon Web Services (AWS)** Santa Clara, United States
Automated Reasoning Group 2025/06 – 2025/09
Built a tool from scratch for validating SMT-LIB arithmetic queries and models: integrated a Rust parser to construct Lean objects, developed a Rust–Lean FFI pipeline, encoded SMT terms and semantics in **Lean**, and formally proved correctness of optimized evaluation against reference semantics.

RESEARCH EXPERIENCE

- **PhD Student at Center for Automated Reasoning, Stanford University** Stanford, United States
Under Prof. Clark Barrett 2024/01 – Present
 1. Working on using **LLMs** to learn programmatic features from **verification** benchmarks to tune solver/verifier heuristics, bridging neural and **formal reasoning** for more adaptive and interpretable verification.
 2. Implemented SMT query normalization in C++, enhancing performance **stability** of **cvc5** and **Z3** across query mutations for all SMT-LIB theories, including industrial **program verification** benchmarks.
- **Research Intern at Dependable Systems Lab, EPFL** Lausanne, Switzerland
Under Prof. George Candea 2022/07 - 2022/08
Improved **PIX** by extending **KLEE's** symbolic execution engine with **Z3's** quantified constraint solving, implementing **loop summarization** to mitigate **path explosion** and scale analysis of **network software**. Involved substantial **C++ systems programming** and **solver integration engineering**.

- **Research Intern at Automated Program Reasoning Group, TU Wien** Vienna, Austria
Under Prof. Laura Kovács and Prof. Ezio Bartocci 2021/07 - 2022/02 + 2023/05 - 2023/12
 1. Extended state-of-the-art techniques for generating **polynomial loop invariants**, enabling support for a broader class of loops. Implemented in Polar, a **static analysis** tool for **reasoning** about **probabilistic programs** using **symbolic** approaches.
 2. Implemented a recursive **program synthesis** framework based on mathematical induction over algebraic datatypes, into Vampire, a state-of-the-art first-order **theorem prover** written in C++.

HONORS AND AWARDS

- **Radhia Cousot Young Researcher Best Paper Award** 2022
29th Static Analysis Symposium (SAS 2022)
- **Stanford School of Engineering (SoE) Fellowship** 2024
Awarded a \$12900/quarter stipend and full tuition coverage for one year
- **Ranked 8th in ACM-ICPC (International Collegiate Programming Contest)** 2020
West Asia Region, Tehran site
- **Summer@EPFL Fellowship** 2022
Ranked top 1.5% among 4000 applicants and awarded a 1600 CHF/month fellowship over summer

PROJECTS

- **SMT model validator** Rust, Lean
Built a tool from scratch for validating models produced by SMT solvers for satisfiable benchmarks.
- **cvc5** —  github.com/cvc5/cvc5 C++
Implemented a normalization pre-processing pass in this industrial-strength SMT solver for achieving performance stability across query mutations.
- **KLEE** —  github.com/bolt-perf-contracts/klee/pull/9 C++, Z3
Integrated Z3's quantified reasoning into KLEE's symbolic execution engine to summarize loops and mitigate path explosion, improving scalability on complex programs involving heavy string operations.
- **ARM Processor Pipeline** —  github.com/daneshvar-amrollahi/ARM Verilog
Implemented a comprehensive 5-stage pipelined ARM processor with data forwarding, hazard detection, cache memory, and SRAM controller.
- **Vampire** —  github.com/vprover/vampire C++
Implemented a framework within this theorem prover for synthesizing verified recursive programs.
- **Polar** —  github.com/probing-lab/polar Python, SymPy
Implemented a polynomial loop-invariant synthesizer for (probabilistic) unsolvable loops.
- **Koloocheh** —  github.com/daneshvar-amrollahi/Koloocheh Python, gRPC
A peer-to-peer file-sharing system employs the flooding algorithm for search operations and ensures a small graph diameter by leveraging random graph properties.
- **xv6** —  github.com/daneshvar-amrollahi/os-lab-xv6 C
Extended xv6 operating system kernel with several extra features such as various new system calls, multilevel queue scheduling, and synchronization.
- **FTP Server** —  github.com/daneshvar-amrollahi/FTP-Server C++, Posix Threads
A multithreaded client-server implementation of an FTP Server using sockets for communication.

SKILLS

- **Programming Languages:**
 - Experienced in C, C++, Python.
 - Familiar with Scala, Rust, Go, Bash, SystemVerilog.
- **Tools:** cvc5, Z3, Lean, KLEE, L^AT_EX.